

# CAR RENTAL REQUEST AUTOMATION PROJECT

## Project Timeline & Milestones:

| Phase   | Activities                                                        | Timeline | Status    |
|---------|-------------------------------------------------------------------|----------|-----------|
| Phase 1 | Project setup and ServiceNow configuration                        | 2 Days   | Completed |
| Phase 2 | Car Rental Request Catalog Item, variables and form configuration | 3 Days   | Completed |
| Phase 3 | Flow Designer automation, approval and catalog task configuration | 3 Days   | Completed |
| Phase 4 | Testing, assignment, notifications and refinement                 | 2 Days   | Completed |
| Phase 5 | Documentation, demo video and final presentation                  | 5 Days   | Completed |

## 1. Troubleshooting

During the development and testing of the Car Rental Request Automation project, several configuration and workflow issues were identified and resolved.

### Issues Identified and Resolutions

| Issue                                                         | Resolution                                                                                                  |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Approval was showing as Cancelled for an earlier test request | A fresh Requested Item (RITM) was created for final testing instead of using the cancelled request          |
| Flow execution was showing some steps as “Not Run”            | The flow conditions and approval path were reviewed to ensure the approved branch was executed              |
| Catalog Task was not progressing in an earlier test           | The approval condition and flow execution sequence were checked before performing the final test            |
| Email notification was not triggered in an earlier test       | The email action and requester email variable were reviewed in the flow                                     |
| Different ServiceNow UI options appeared during configuration | The available options in the current ServiceNow instance were used to complete the equivalent configuration |

## **Troubleshooting Approach**

The troubleshooting process involved:

1. Reviewing the Flow Designer execution details.
2. Checking the approval status of the Requested Item (RITM).
3. Verifying the If Approved condition.
4. Checking the Create Catalog Task action.
5. Verifying the Get Catalog Variables action.
6. Checking the Send Email action.
7. Creating a fresh test request after resolving earlier test issues.
8. Verifying the complete workflow from request submission to notification.

## **Result**

The troubleshooting process helped identify configuration and testing issues and ensured that the workflow could be validated systematically before the final project demonstration.

## **2. Adherence to Timelines**

The project was developed in a structured phase-wise manner. Each phase was planned with specific activities and milestones. The Car Rental Request catalog item, variables, Flow Designer automation, approval process and catalog task assignment were completed progressively. Testing and troubleshooting were performed before moving to the final documentation and presentation phase.

## **3. Innovation**

Innovation in the Car Rental Request Automation project focuses on reducing manual processing by using ServiceNow Service Catalog and Flow Designer to automate request submission, approval, task assignment and notifications.

### **Innovative Features:**

#### **a. Automated Approval Workflow**

The system automatically initiates an approval process when a user submits a car rental request. This reduces manual intervention and ensures that every request follows a standardized approval process.

#### **b. Dynamic Request Data**

The workflow uses information entered by the requester, such as requested date, pickup location, drop location, duration and car type. This allows the workflow to process each request according to the information provided by the user.

### **c. Automatic Task Assignment**

After approval, the system automatically creates a Catalog Task and assigns it to the Transport Team. This eliminates the need for manual task creation and assignment.

### **d. Automated Email Notification**

The system sends an automated email notification after the request is approved, keeping the requester informed about the status of the car rental request.

### **e. End-to-End Automation**

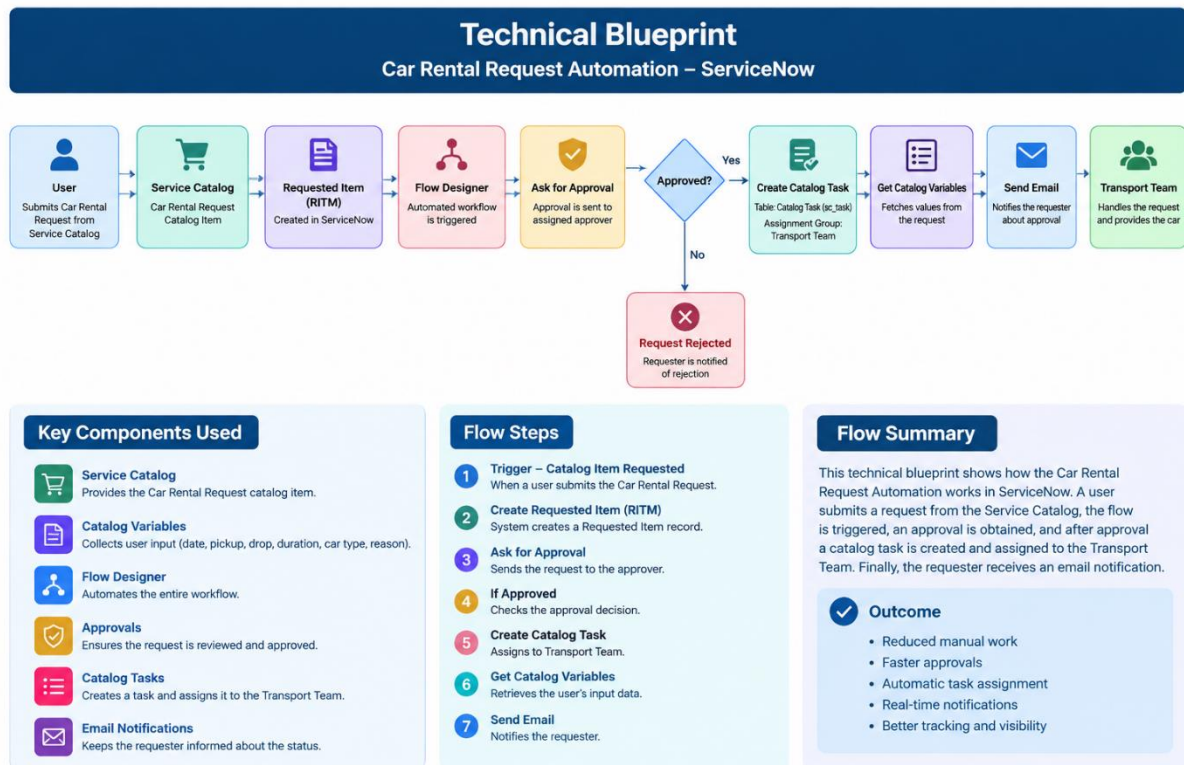
The project provides an end-to-end automated process starting from request submission and continuing through approval, task creation, assignment and notification. This improves efficiency, consistency and request tracking.

| <b>Innovation</b>                   | <b>Benefit</b>                          |
|-------------------------------------|-----------------------------------------|
| Automated approval                  | Reduces manual approval handling        |
| Dynamic catalog data                | Processes requests based on user input  |
| Automatic task creation             | Eliminates manual task creation         |
| Automatic Transport Team assignment | Sends work to the appropriate team      |
| Email notification                  | Keeps requester informed                |
| End-to-end workflow                 | Provides a consistent automated process |

### **Innovation Summary:**

The project demonstrates how ServiceNow can transform a manual car rental request process into an automated workflow. By combining Service Catalog, Flow Designer, approvals, Catalog Tasks and email notifications, the solution improves efficiency, reduces manual effort and provides better visibility into the request lifecycle.

## 4. Document Technical Blueprint



## 5. Setup Manual

### Car Rental Request Automation — ServiceNow

This setup manual describes the steps required to configure and implement the Car Rental Request Automation project in a ServiceNow instance.

#### a. System Requirements

##### Prerequisites

- ServiceNow Developer Instance
- Access to Service Catalog
- Access to Flow Designer
- Appropriate permissions to create catalog items and flows
- User records for requester and approver
- Transport Team assignment group

#### b. Create the Catalog Item

##### Step 1 — Create Car Rental Request Catalog Item

1. Navigate to All → Service Catalog → Catalog Definitions → Maintain Items.
2. Click New.

3. Enter the catalog item name:  
Car Rental Request
4. Configure the catalog item.
5. Add the required variables.

### c. Configure Catalog Variables

| Variable         | Purpose                                          |
|------------------|--------------------------------------------------|
| Email            | Email address of the requester for notifications |
| Requester        | Identifies the person requesting the rental      |
| Requested Date   | Specifies the required rental date               |
| Pickup Location  | Location where the car is picked up              |
| Drop Location    | Destination                                      |
| Duration (Hours) | Specifies rental duration                        |
| Car Type         | Specifies the requested vehicle type             |
| Reason           | Explains why the car is required                 |

The catalog variables collect the information required to process each car rental request.

### d. Configure Car Type Choices

The Car Type variable contains the available vehicle options, including Mini, SUV, Hatchback, Luxury and Sedan.

### e. Create the Flow

**Step 2** — Create Flow Designer Workflow

Navigate to:

All → Flow Designer

Create/open the flow for:

Car Rental Request

Configure the trigger:

Catalog Item Requested

This starts the automation whenever a user submits the Car Rental Request catalog item.

## **f. Configure Approval**

### **Step 3 — Configure Approval**

Add the Ask for Approval action in Flow Designer.

The request is sent for approval after the catalog request is submitted. The workflow then checks the approval result and determines whether the request is approved or rejected.

## **g. Configure Approved Condition**

### **Step 4 — Check Approval Result**

Add an If condition to check whether the approval was successful.

If Approval = Approved

When the condition evaluates to true, the workflow continues to the next actions.

## **h. Create Catalog Task**

### **Step 5 — Create Catalog Task**

Add:

Create Catalog Task

Configure:

Table: Catalog Task [sc\_task]

Assignment Group: Transport Team

The catalog task represents the work required by the transport team to fulfill the approved request.

## **i. Retrieve Catalog Variables**

### **Step 6 — Get Catalog Variables**

Use the Get Catalog Variables action to retrieve the information entered by the requester.

The workflow can use values such as:

- Requested Date
- Pickup Location
- Drop Location
- Duration
- Car Type
- Reason
- Email
- Requester

## **j. Configure Email**

### **Step 7 — Send Email Notification**

Add:

Send Email

Configure the email to notify the requester after approval.

Example subject:

Your Car Request is approved.

Example message:

Dear,

Your car rental request has been approved.

Regards,

Car Rental Automation System

## **k. Test the Configuration**

### **Step 8 — Test the Workflow**

1. Open the Car Rental Request catalog item.
2. Enter the required information.
3. Submit the request.
4. Open the generated RITM.
5. Verify the approval.
6. Approve the request.
7. Check the Flow Designer execution.
8. Verify that the Catalog Task is created.
9. Verify that it is assigned to Transport Team.
10. Verify that the email notification is sent.

## **l. Expected Result**

When a user submits a Car Rental Request, ServiceNow creates a Requested Item and initiates the automated approval process. After approval, a Catalog Task is automatically created and assigned to the Transport Team. The requester is then notified through email.

# **6. Project Demo Video Planning**

## **Car Rental Request Automation — ServiceNow**

### **Objective:**

The demo video will demonstrate the complete workflow of the Car Rental Request Automation system, from submitting a request through the Service Catalog to approval, automatic task creation, assignment to the Transport Team and email notification.

**Demo sequence:**

| Scene | What to Demonstrate             | What to Explain                                      |
|-------|---------------------------------|------------------------------------------------------|
| 1     | Project Introduction            | Briefly explain the purpose of the project           |
| 2     | Car Rental Request Catalog Item | Show the request form and available fields           |
| 3     | Submit Request                  | Enter sample rental details and submit               |
| 4     | Requested Item (RITM)           | Show the generated request                           |
| 5     | Approval                        | Show the approval process                            |
| 6     | Approve Request                 | Approve the request                                  |
| 7     | Flow Designer Execution         | Show the workflow progressing after approval         |
| 8     | Catalog Task                    | Show the task created and assigned to Transport Team |
| 9     | Email Notification              | Show the approval email                              |
| 10    | Conclusion                      | Summarize the benefits of automation                 |

**Demo script:****Introduction**

"Hello, this is my project, Car Rental Request Automation using ServiceNow. The purpose of this project is to automate the car rental request process, including request submission, approval, task assignment and notification."

## **Catalog Item**

"This is the Car Rental Request catalog item. The requester provides details such as requested date, pickup location, drop location, duration, car type, reason and requester information."

## **Request Submission**

"I will now submit a new car rental request using the required information."

## **Approval**

"After submitting the request, ServiceNow creates a Requested Item and initiates the approval process."

## **Flow**

"Once the request is approved, Flow Designer automatically continues the workflow and creates a catalog task."

## **Assignment**

"The catalog task is automatically assigned to the Transport Team for fulfillment."

## **Notification**

"The system also sends an email notification to keep the requester informed about the approval."

## **Conclusion**

"This automation reduces manual effort, improves request tracking and provides a consistent process for handling car rental requests."

## **Actual recording plan**

Catalog Item → Submit → RITM → Approval → Approved → Flow Execution → Catalog Task → Transport Team → Email

## **Demo Video Completion**

The project demonstration video was successfully recorded with voice narration. The video demonstrates the complete Car Rental Request Automation workflow, including request submission, Requested Item (RITM) creation, approval process, Flow Designer execution, catalog task assignment to the Transport Team, and email notification.

# **7. Visual Demonstration**

The visual demonstration provides screenshots of the key components and workflow stages of the Car Rental Request Automation project. These screenshots demonstrate the configuration, execution and outcome of the automated process in ServiceNow.

| No. | Screenshot / Visual Evidence    | Purpose                                        |
|-----|---------------------------------|------------------------------------------------|
| 1   | Car Rental Request Catalog Item | Shows the request form and available fields    |
| 2   | Catalog Item Variables          | Shows the configured request variables         |
| 3   | Car Type Choices                | Shows available car types                      |
| 4   | Flow Designer                   | Shows the complete automation workflow         |
| 5   | Approval Configuration          | Shows the approval action                      |
| 6   | Requested Item (RITM)           | Shows the submitted rental request             |
| 7   | Approved Request                | Shows the successful approval                  |
| 8   | Catalog Task                    | Shows the task assigned to Transport Team      |
| 9   | Flow Execution                  | Shows the successful execution of the workflow |
| 10  | Email Notification              | Shows the notification sent to the requester   |

### Visual Evidence:

Screenshots of the key stages of the Car Rental Request Automation project will be captured from the ServiceNow instance to demonstrate the configuration and execution of the workflow.

## 8. Explanation Clarity

The project is designed to provide a clear and structured explanation of the Car Rental Request Automation process. The system workflow, ServiceNow components and automation steps can be explained clearly during the project demonstration.

### Project Objective

The objective of the project is to automate the car rental request process using ServiceNow.

### User Role

The requester submits a Car Rental Request through the Service Catalog by providing the required rental details.

### **Approval Role**

The approver reviews the submitted request and approves or rejects it.

### **Fulfillment Role**

After approval, a Catalog Task is automatically created and assigned to the Transport Team for further processing.

### **Automation**

Flow Designer automates the approval, task creation, assignment and notification process.

### **Notification**

The requester receives an email notification regarding the approval of the car rental request.

### **Workflow**

Requester  
↓  
Submit Car Rental Request  
↓  
RITM Created  
↓  
Approval  
↓  
Approved  
↓  
Catalog Task Created  
↓  
Transport Team  
↓  
Email Notification

### **Conclusion**

The project demonstrates a clear end-to-end automated process in ServiceNow, with defined responsibilities for the requester, approver and fulfillment team.

## **9. Realism & Quality**

### **Introduction**

The Car Rental Request Automation project is designed to represent a realistic organizational car rental request process. The solution uses ServiceNow components to provide a structured and automated workflow from request submission to fulfillment.

### **Realistic Request Form**

The catalog item collects practical information required for a car rental request, including requester, requested date, pickup location, drop location, duration, car type and reason.

### **Realistic Approval Process**

The request is submitted for approval before the rental task is processed, representing a controlled organizational approval process.

### **Automatic Task Assignment**

Once the request is approved, a catalog task is automatically created and assigned to the Transport Team for fulfillment.

### **Automated Notification**

Email notification keeps the requester informed about the status of the request.

### **Workflow Consistency**

Flow Designer provides a standardized process so that requests follow the same sequence of approval, task creation and notification.

## **Quality Checklist**

| <b>Quality Area</b>                  | <b>Status</b> |
|--------------------------------------|---------------|
| Catalog item configured              | Complete      |
| Required variables configured        | Complete      |
| Car type choices configured          | Complete      |
| Approval workflow configured         | Complete      |
| Catalog task configured              | Complete      |
| Transport Team assignment configured | Complete      |
| Email notification configured        | Complete      |
| Workflow tested                      | Complete      |

### **Quality Summary:**

The project combines realistic request information with automated ServiceNow workflows to provide a structured and consistent car rental request process. The implementation is designed to reduce manual processing and improve request tracking.

## **10. Scalability & Future Plan**

### **Introduction**

The Car Rental Request Automation system can be extended with additional features to support a larger number of users, vehicles and rental requests. The current workflow provides a foundation that can be enhanced to meet future organizational requirements.

### **Future Enhancements**

#### **a. Automatic Car Availability**

The system can be enhanced to automatically check vehicle availability before processing a rental request.

#### **b. Automatic Price Calculation**

A future version can calculate the rental cost based on car type, rental duration and other applicable charges.

#### **c. Multi-Level Approval**

Additional approval levels can be introduced for high-value or long-duration rental requests.

#### **d. Car Assignment**

The system can automatically assign an available vehicle after the request is approved.

#### **e. Cancellation and Modification**

Users could be provided with options to modify or cancel their rental requests before fulfillment.

#### **f. Dashboard and Reporting**

A ServiceNow dashboard could display information such as total requests, pending approvals, approved requests, rejected requests and completed rentals.

#### **g. Mobile Access**

The solution could be extended to provide convenient access to requests and approvals through mobile devices.

| <b>Future Enhancement</b>     | <b>Purpose</b>                                   |
|-------------------------------|--------------------------------------------------|
| Car availability checking     | Verify vehicle availability                      |
| Automatic price calculation   | Calculate rental cost                            |
| Multi-level approval          | Handle different approval requirements           |
| Automatic car assignment      | Assign available vehicles                        |
| Cancellation and modification | Provide greater request flexibility              |
| Dashboard and reporting       | Monitor rental activity                          |
| Mobile access                 | Support requests and approvals on mobile devices |

#### **Future Scope:**

The proposed enhancements can make the Car Rental Request Automation system more flexible, scalable and suitable for larger organizational requirements while building on the existing ServiceNow workflow.

# PROJECT CONCLUSION

## Conclusion

The Car Rental Request Automation project was successfully developed using ServiceNow to automate the car rental request process. The system provides a structured process for users to submit car rental requests through the Service Catalog and enables automated approval, task assignment, and notification.

The project uses Flow Designer to automate the workflow from request submission to approval and fulfillment. Once a request is approved, a catalog task is created and assigned to the Transport Team for further processing. Email notifications are also used to keep the requester informed about the request status.

The project was tested to verify the configured workflow and its major components. The complete project demonstration was also recorded with voice narration to explain the implemented process and functionality.

Overall, the project demonstrates how ServiceNow can be used to reduce manual effort, improve request tracking, automate approvals and task assignments, and provide a consistent process for managing car rental requests.

## Project Outcome

- Car Rental Request Catalog Item created
- Request variables configured
- Car Type choices configured
- Approval workflow implemented
- Flow Designer automation completed
- Catalog Task created and assigned to Transport Team
- Email notification configured
- Workflow tested
- Project documentation completed
- Demo video recorded with voice narration

## Future Enhancements

The system can be further enhanced with:

- Automatic car availability checking
- Automatic price calculation
- Automatic car assignment

- Multi-level approval
- Request cancellation and modification
- Dashboard and reporting
- Mobile access

**Final Project Status :** Completed